MATH INQUIRERS HOLIDAY PROGRAM

Free 30-minute taster

Lesson 1: Number Sleuths (Numbers to 10 000) • Math Inquirers Level 3 (ages 8-10)

Notice it. Test it. Explain why.

What this is

A short sample of a real Math Inquirers lesson for Level 3 (ages 8-10, covers P3 topics). It is Lesson 1 of the Level 3 holiday program (Number Sleuths), shortened to about 30 minutes: a warm-up, a hands-on explore, a mini-lesson, a few worksheet questions and an exit ticket. The same routine runs through all ten 90-minute lessons.

The 30 minutes

Time	Part	What happens
5 min	**Warm-up**	Biggest number wins: the tutor draws four digits one at a time; children write each one in Th, H, T or O straight away and cannot move it. The biggest number scores a point. Then play for the smallest.
8 min	**Explore**	Pairs use the cards 2, 4, 6 and 8 to make every 4-digit number between 4000 and 6000, then find the one closest to 5000.
5 min	**Mini-lesson**	Greatest and smallest: decide the thousands digit first (0 can never go first). Odd or even: choose the ones digit first. Closest: check the nearest number above and the nearest below. Make a list in order: fix the first digit, then list the rest from smallest.
9 min	**Worksheet**	The child works on questions 1 to 4 of the Lesson 1 sheet (next pages), then checks them with the tutor.
3 min	**Exit ticket**	Use the cards 6, 0, 3, 8 once each. Write the smallest 4-digit number and the greatest odd 4-digit number.

What you need

Digit cards 0–9 (two sets per child), a place value mat (Th H T O), mini whiteboards, a 10-sided die or a bag of digit cards 0–9 for the warm-up game. For the sneak-peek puzzles: paper, a pencil and a few counters or small objects for each child.

Online: Children cut out the digit cards at the back of the workbook (or write 0–9 on small paper squares) and draw a Th H T O mat on paper. The tutor draws the digits on camera.

Pages in this file: this page, the child's worksheet (about two pages), then tutor notes with the answers. Do not give the tutor-notes pages to the child. This taster is enrichment for thinking skills; it does not promise exam results and is not affiliated with MOE or the Gifted Education Programme.

NUMBER SLEUTHS

Name _____ Group _____ Date _____

Digit cards: use each card once in each number. A 4-digit number cannot start with 0.

No.1

Use these digit cards once each to make 4-digit numbers.

3	0	7	5

a The greatest number: _____

b The smallest number: _____

c The greatest ODD number: _____

d The number closest to 5000: _____

e Why is (d) the closest? It is only _____ away from 5000.

No.2

Make a list in order. Write ALL the 4-digit numbers you can make with the cards 2, 0, 5, 5. Start with the smallest.

Starting with 2	
Starting with 5	

a How many numbers did you find? _____

b How many of them are greater than 5200? _____

No.3

Look for a pattern. Write the next two numbers.

a 9850, 9750, 9650, _____, _____

b 1234, 2345, 3456, _____, _____

c 1000, 1001, 1003, 1006, 1010, _____, _____

d 3000, 3250, 3500, 3750, … Keep going. Will 5000 be in the pattern? _____
Which number comes right after 5000? _____

No.4

Place value puzzles.

Hint: 10 hundreds make 1 thousand. You can draw a box for each thousand, hundred, ten and one.

a 4 thousands, 15 hundreds, 2 tens and 7 ones make _____

b 7305 = 6 thousands + _____ hundreds + 5 ones

c Mei says that 6 thousands, 0 hundreds and 23 tens is 6023. Is she right? Explain.

Tutor notes: Lesson 1, Number Sleuths

Numbers to 10 000 • Level 3 (ages 8-10, covers P3 topics)

Goal

Read and build 4-digit numbers by place value; make the greatest, smallest, odd, even and closest numbers from digit cards; list every number in order; solve clue puzzles and continue number patterns.

Skills: Place value to 10 000, renaming (15 hundreds = 1 thousand 5 hundreds), making a list in order, looking for a pattern, using clues one at a time, explaining "I have them all"

Opening

Welcome, Number Sleuths! Every number hides clues in its digits. Today we use the clues to find mystery numbers, build the biggest and the smallest, and prove that we have found every number.

Warm-up (about 5 minutes)

Biggest number wins: the tutor draws four digits one at a time; children write each one in Th, H, T or O straight away and cannot move it. The biggest number scores a point. Then play for the smallest.

Draw a Th H T O mat. I will call out four digits, one at a time. Write each digit in a box straight away. No moving!

Who made the biggest number? Where did you put the 9? Why there?

Explore (about 8 minutes)

Pairs use the cards 2, 4, 6 and 8 to make every 4-digit number between 4000 and 6000, then find the one closest to 5000.

Predict: With the cards 2, 4, 6 and 8, how many 4-digit numbers between 4000 and 6000 can you make? Write a guess first.

- Stuck: “Which card must go in the thousands place if the number is between 4000 and 6000?”
- Missing some: “Keep the 4 in front. Now change the hundreds digit in order: 2, then 6, then 8.”
- Ready for more: the Go further what-if on the slide (cards 2, 4, 5, 8: 12 numbers between 4000 and 6000). Or: “How many numbers can you make altogether with the four cards?” (24: six start with each card.)

Mini-lesson (about 5 minutes)

Greatest and smallest: decide the thousands digit first (0 can never go first). Odd or even: choose the ones digit first. Closest: check the nearest number above and the nearest below. Make a list in order: fix the first digit, then list the rest from smallest.

Worksheet answers (questions 1 to 4)

Q1	(a) 7530 (b) 3057 (c) 7503 (d) 5037 (e) It is only 37 away from 5000. The closest number below 5000 is 3750, which is 1250 away.
Q2	Starting with 2: 2055, 2505, 2550. Starting with 5: 5025, 5052, 5205, 5250, 5502, 5520. (a) 9 numbers (b) 4 (5205, 5250, 5502, 5520)

Q3	(a) 9550, 9450 (take away 100) (b) 4567, 5678 (add 1111) (c) 1015, 1021 (add 1, 2, 3, 4, then 5, then 6) (d) Yes, 5000 is in the pattern (3000 + 8 jumps of 250). The number right after it is 5250
Q4	(a) 5527 (b) 13 hundreds (c) No. 23 tens is 230, so the number is 6000 + 230 = 6230.

Talk time

Mei says the smallest 4-digit number she can make with 3, 0, 7, 5 is 0357. Do you agree?

Answer: No. 0357 is really 357, a 3-digit number, so the smallest 4-digit number is 3057. We know a list is complete because we fixed the first digit and listed the rest in order from smallest, so nothing is missed or repeated.

Exit ticket (about 3 minutes)

Use the cards 6, 0, 3, 8 once each. Write the smallest 4-digit number and the greatest odd 4-digit number.

Answer: Smallest: 3068. Greatest odd: 8603 (3 is the only odd card, so it goes in the ones place).

Watch for these mistakes

- Putting 0 first for the smallest number (0357 is really 357, a 3-digit number).
- For the greatest odd number, making the greatest number first and then trying to fix it, instead of choosing the ones digit first.
- Listing numbers in a random order, so some are missed or written twice (especially with the two 5s in Q2).

Extension for fast finishers

With the cards 1, 2, 3, 4, how many 4-digit numbers can you make? How many of them are odd? (24 numbers; 12 are odd, because 6 end in 1 and 6 end in 3.)

After the taster

Tell the parent one thing their child noticed and one thing they explained well. Then share the next steps: [dates, fee and sign-up link]. The free diagnostic for the matching level helps choose between Level 3 (ages 8-10, covers P3 topics) and the level either side (see the placement note).